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INTRODUCTION

In today's fast-paced corporate and retail environments, data-driven decisions separate thriving business entities from stagnating ones. This project focuses on analysing a comprehensive corporate revenue sales dataset to uncover hidden trends, evaluate product performance, and optimize sales strategies. By leveraging a modern data stack—combining the structural flexibility of a NoSQL database with the analytical power of Power BI —this project transforms raw transactional logs into actionable business intelligence.

The primary business goals of this analysis are to solve specific operational friction points and identify key growth vectors. Specifically, the analysis is engineered to identify revenue drivers by determining which product categories and regions generate the highest profit margins. Furthermore, looking into historical data allows the business to analyse seasonality, pinpointing peak buying trends to help inventory managers optimize stock levels and prevent costly stockouts. Finally, the project handles customer segmentation, highlighting high-value customer segments to assist marketing teams in refining targeted campaigns and improving overall Company Lifetime Value.

Unlike traditional data projects that rely solely on flat CSV files or rigid relational databases, this system implements a dynamic, semi-structured data pipeline. Document-based records are securely housed within a MongoDB environment, allowing for nested customer information and variable transaction lengths. This architecture drastically reduces local memory overhead and ensures that data processing scales efficiently before advanced statistical analysis and visualization take place.

DATASET DESCRIPTION

The dataset used throughout this project contains historical transaction records for a global enterprise operating in both consumer and corporate markets. Because individual sales records inherently include nested attributes—such as detailed customer profiles and itemized arrays of purchased products—MongoDB was selected as the optimal storage engine. This document-based database format allows the dataset to scale horizontally without the rigid design constraints or intensive join operations typical of a traditional relational schema.

Each transaction is captured as a binary JSON document containing specific data fields that map out the lifecycle of a sale. Every record is uniquely identified by a default MongoDB object identifier, which sits alongside an alpha-numeric transaction sequence number and an exact timestamp detailing when the purchase occurred. The nested customer object breaks down individual buyers by their market segment, such as corporate or individual consumers, as well as their geographic region. The transactional core of the document consists of an array of items, where each element explicitly tracks the product identifier, its parent department category, the quantity of units purchased, and the unit price. Final fields record the payment method utilized and the overall fulfilment status of the order.

The structure of the data can be conceptually organized into a data dictionary that governs how the fields are treated during analysis. The raw transaction ID and order status fields function as categorical indicators, allowing for rapid filtering of valid sales revenue. Temporal trends are extracted directly from the system timestamp to map out from monthly, yearly revenue cycles. Meanwhile, numerical metrics like the quantity and unit price fields serve as the foundation for all quantitative variables, as they are multiplied dynamically during the query phase to calculate gross margins and overall corporate revenue.

IMPLEMENTATION

The implementation of this project based on heavy data aggregation tasks, which were pushed entirely to the database layer. Instead of executing a resource-heavy import of millions of raw database documents into local memory, a custom JSON aggregation pipeline was deployed directly within MongoDB. This pipeline begins by filtering out unfulfilled transactions to ensure data integrity. It then flattens the nested arrays of purchased items, groups the remaining records by their department categories, and calculates the total revenue and units sold simultaneously before sorting the final output in descending order.

By establishing a secure client connection to the database instance, the script injects the JSON aggregation pipeline payload natively into the collection environment. The aggregation pipelines were there after being analysed and pushed to JSON files.

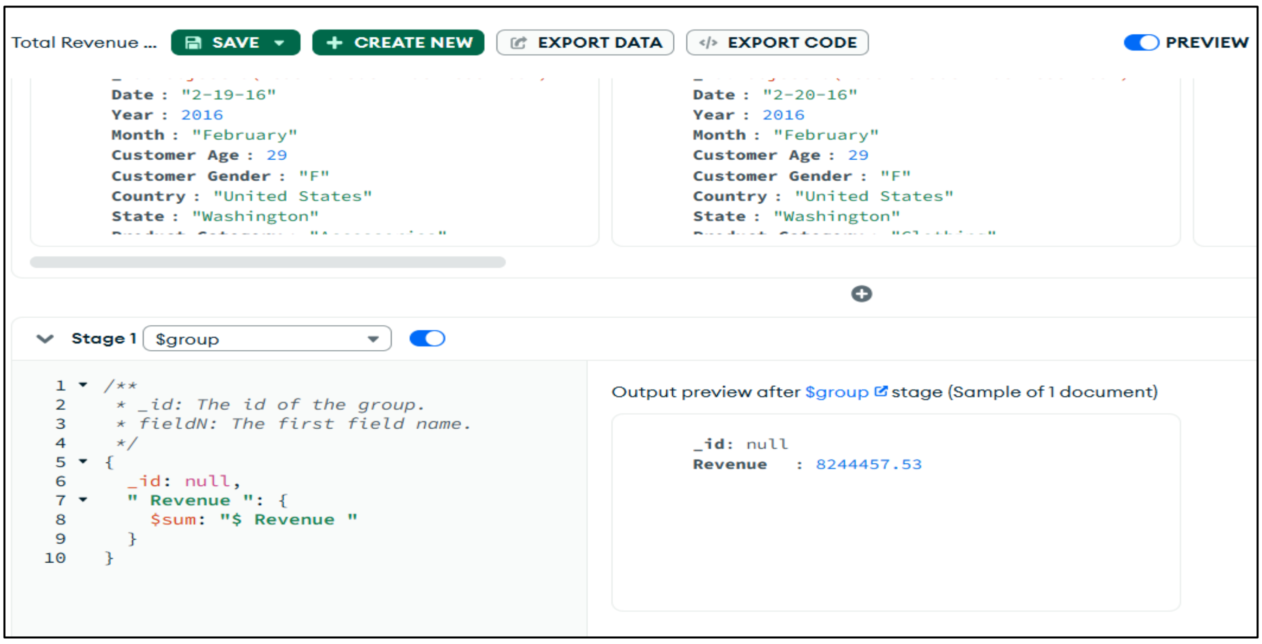
The final stage of implementation focuses on exploratory data analysis and visual communication. Utilizing advanced visualization tools, like Microsoft Power BI, the processed data was used to generate clean visual assets, such as bar charts representing top revenue categories and line graphs illustrating seasonal revenue trends.

These visual outputs directly show the execution of the aggregation pipelines, proving that raw, unstructured transactional documents can be systematically parsed, aggregated at scale within a NoSQL database, and smoothly translated into polished business insights.

SAMPLE OUTPUTS

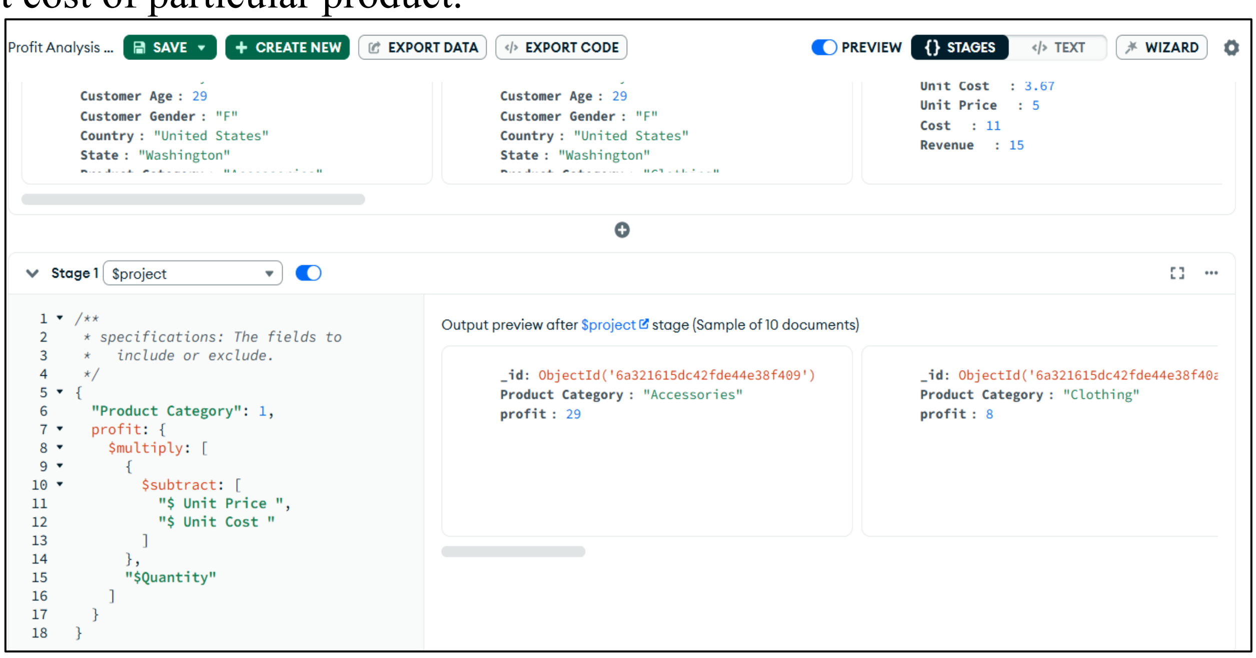
1. Total Revenue Generated by Company:

The analysis provides the overall performance of the company's business from the use of Revenue Sales Dataset. It shows the total revenue generated by the company.



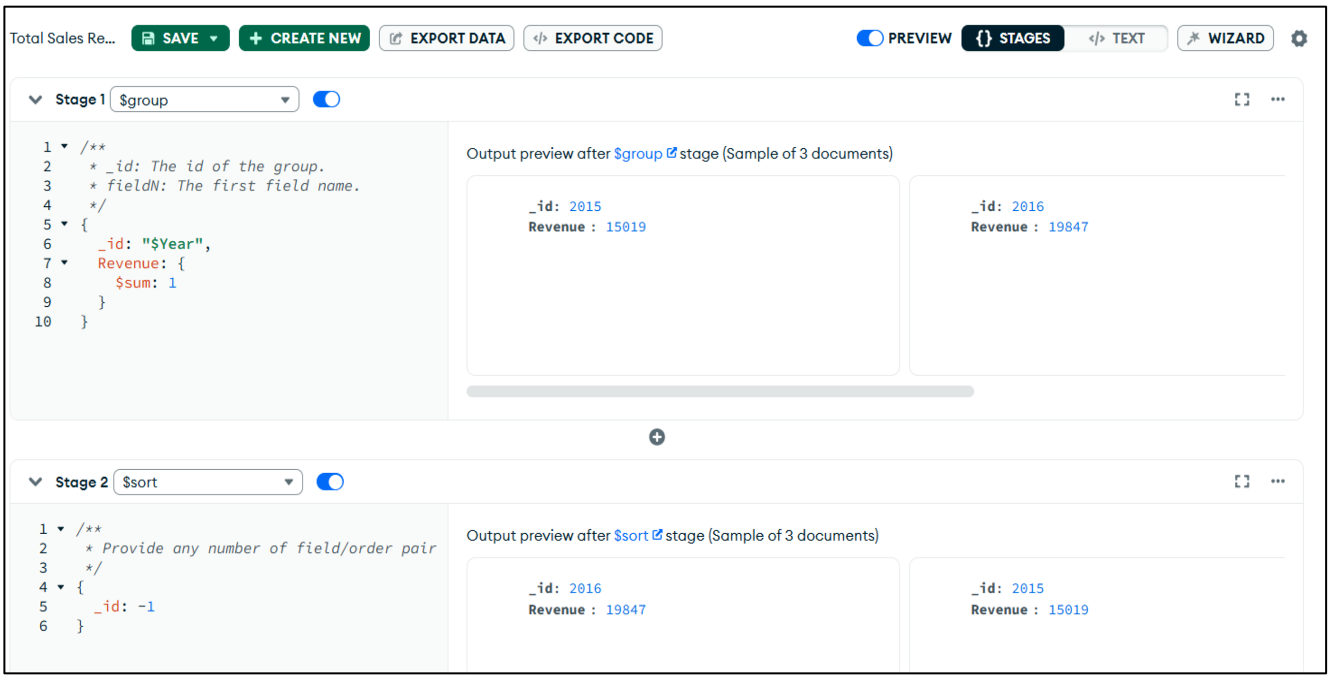
2. Profit Analysis by Category:

This aggregation analysis includes the output of profit generated by the product category section of the dataset, which works by subtracting the unit price to the unit cost of particular product.



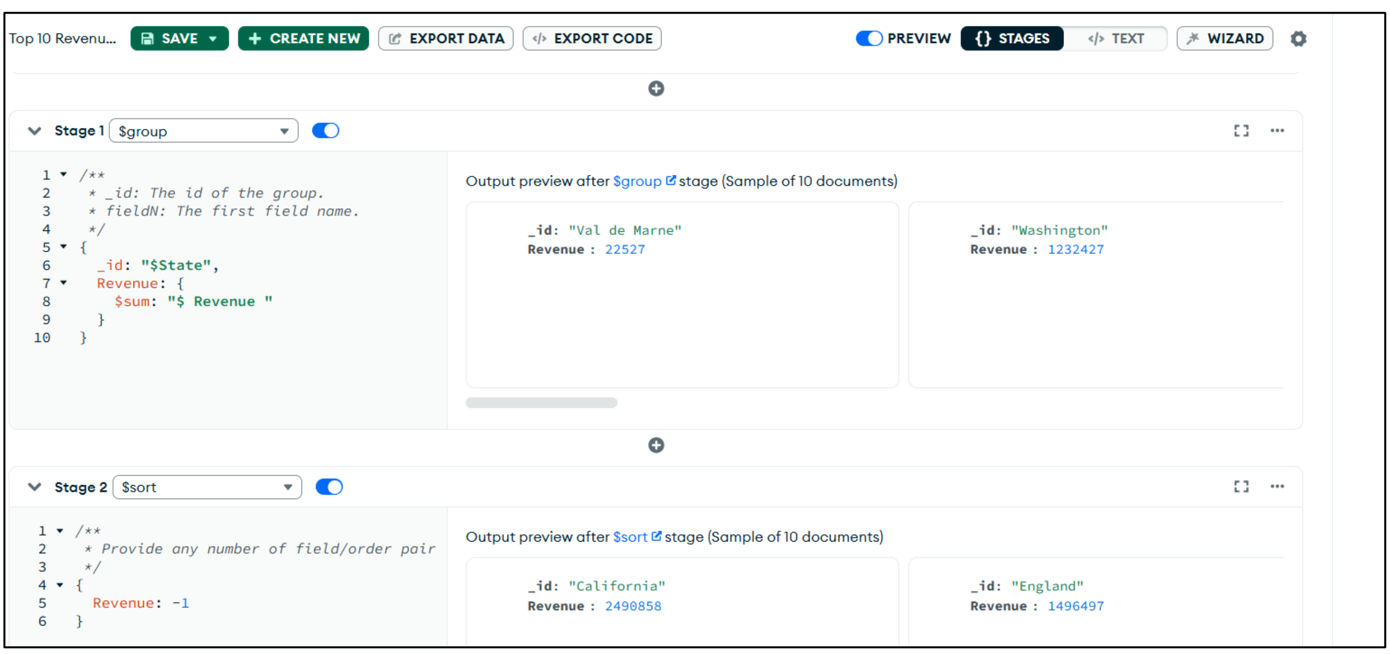
3. Total Sales Revenue by Year:

This output demonstrates the distribution of total sales in that particular year that is, 2015 or 2016.



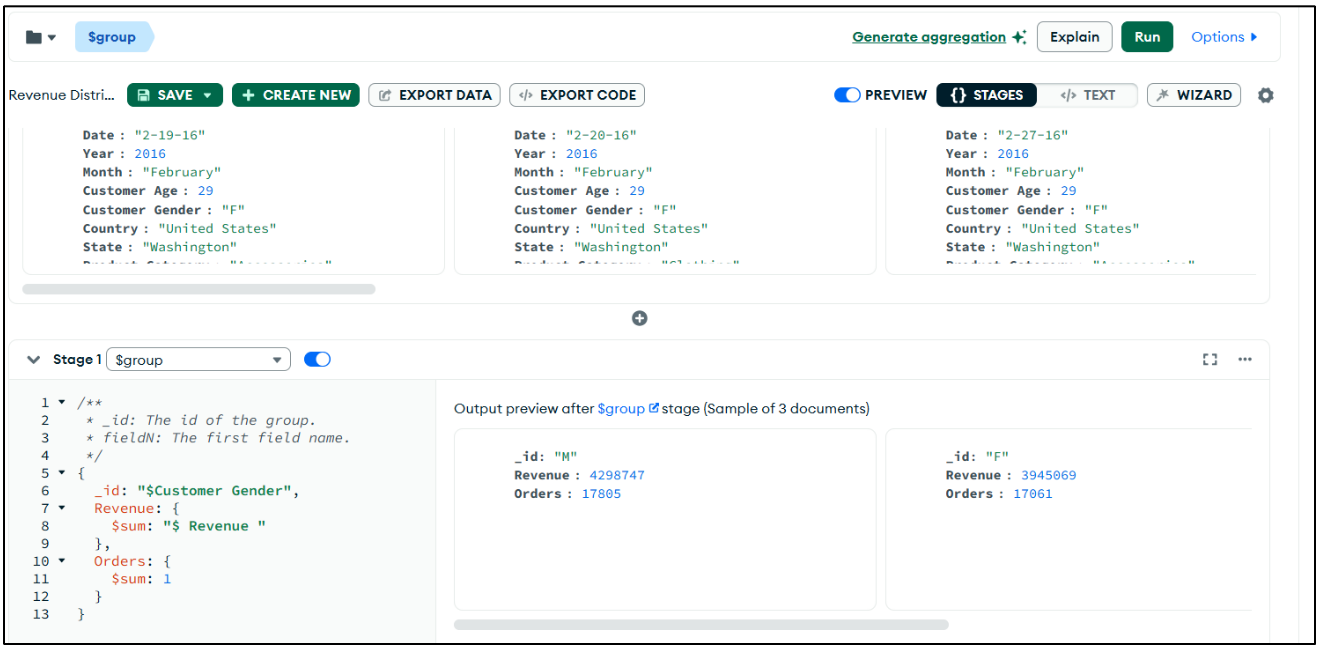
4. Top 10 Revenue generating states:

This analysis displays how the revenue of the company generated from the states of the particular country will be displayed. And we are considering the top 10 states from the country.



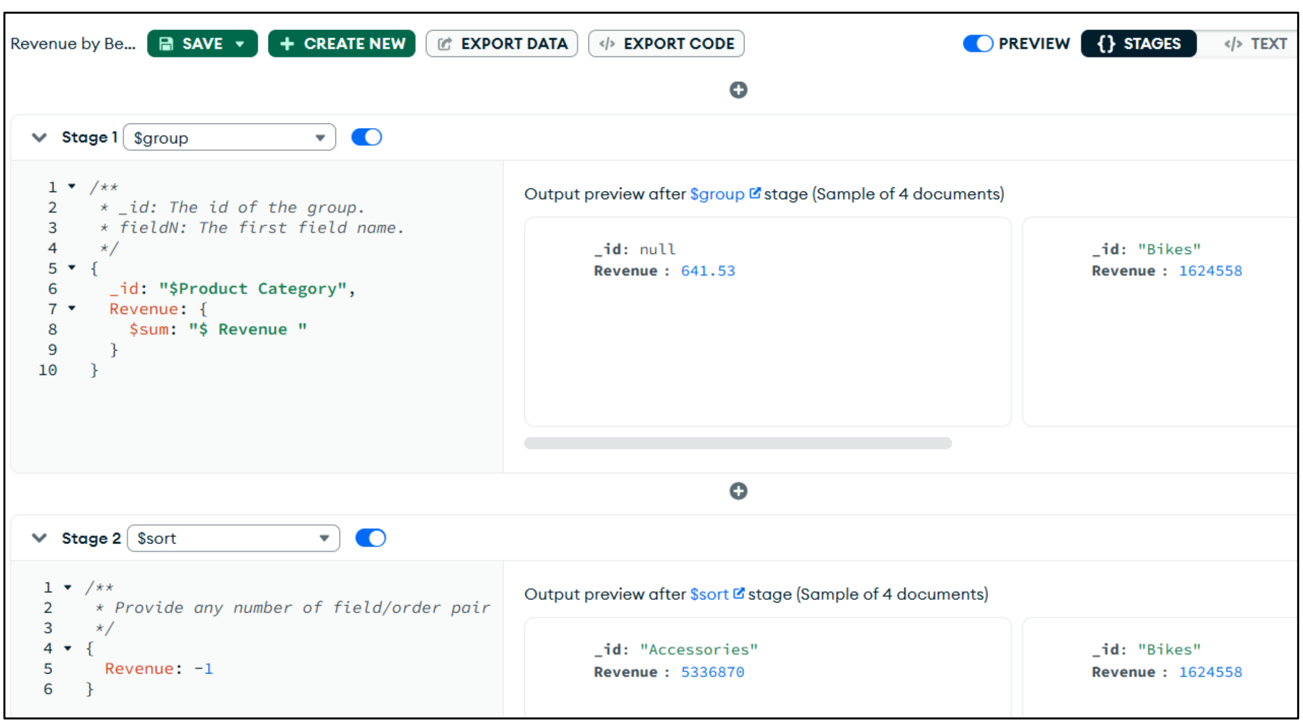
5. Revenue distribution by Gender:

This output shows the distribution of the content by Gender, displaying the revenue distribution of the company in respect to Male(M) or Female(F).



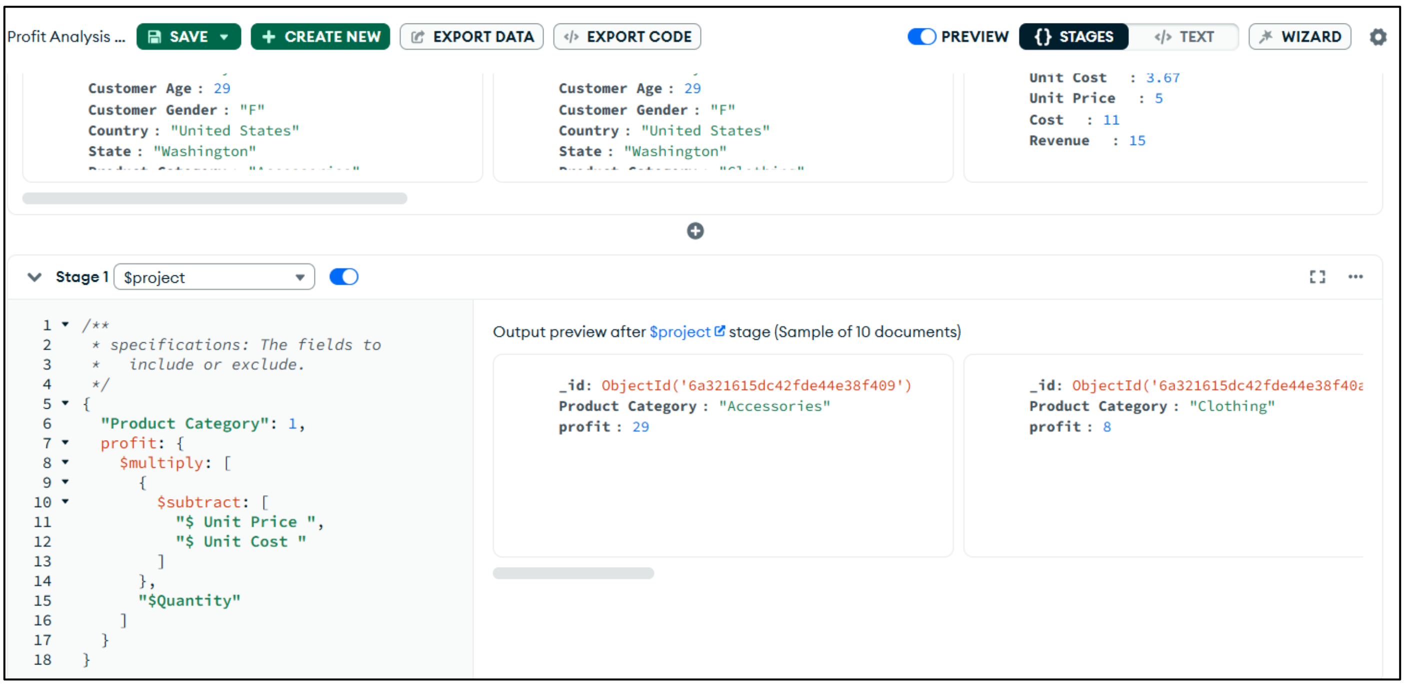
6. Revenue by best performing product categories:

This demonstrates the product category from the Revenue Sales Dataset and show cases the best 10 performing categories along with the revenue generated from the individual categories from the dataset.



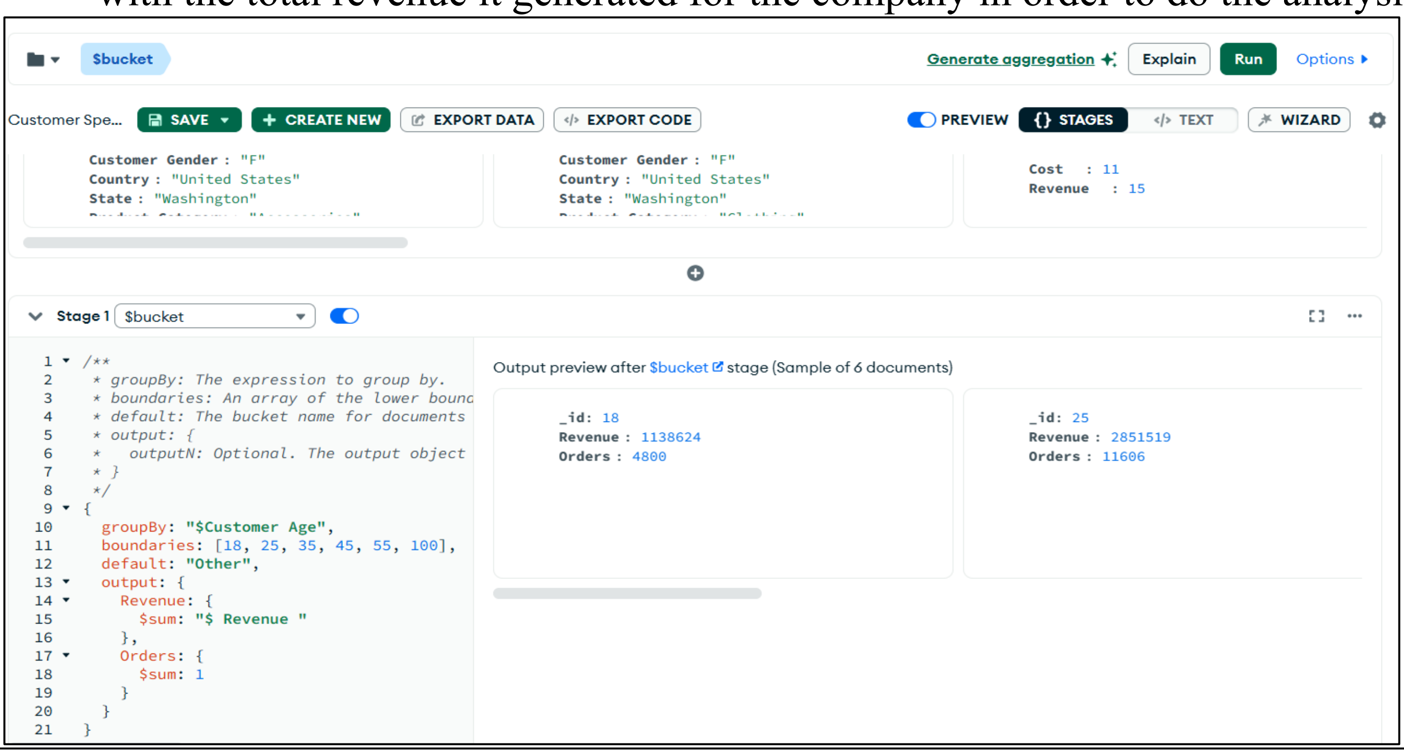
7. Profit Analysis by Product Category:

The analysis focuses on the profit generated from each of the product category which shows the importance and weightage of that particular product in the company, in the market.



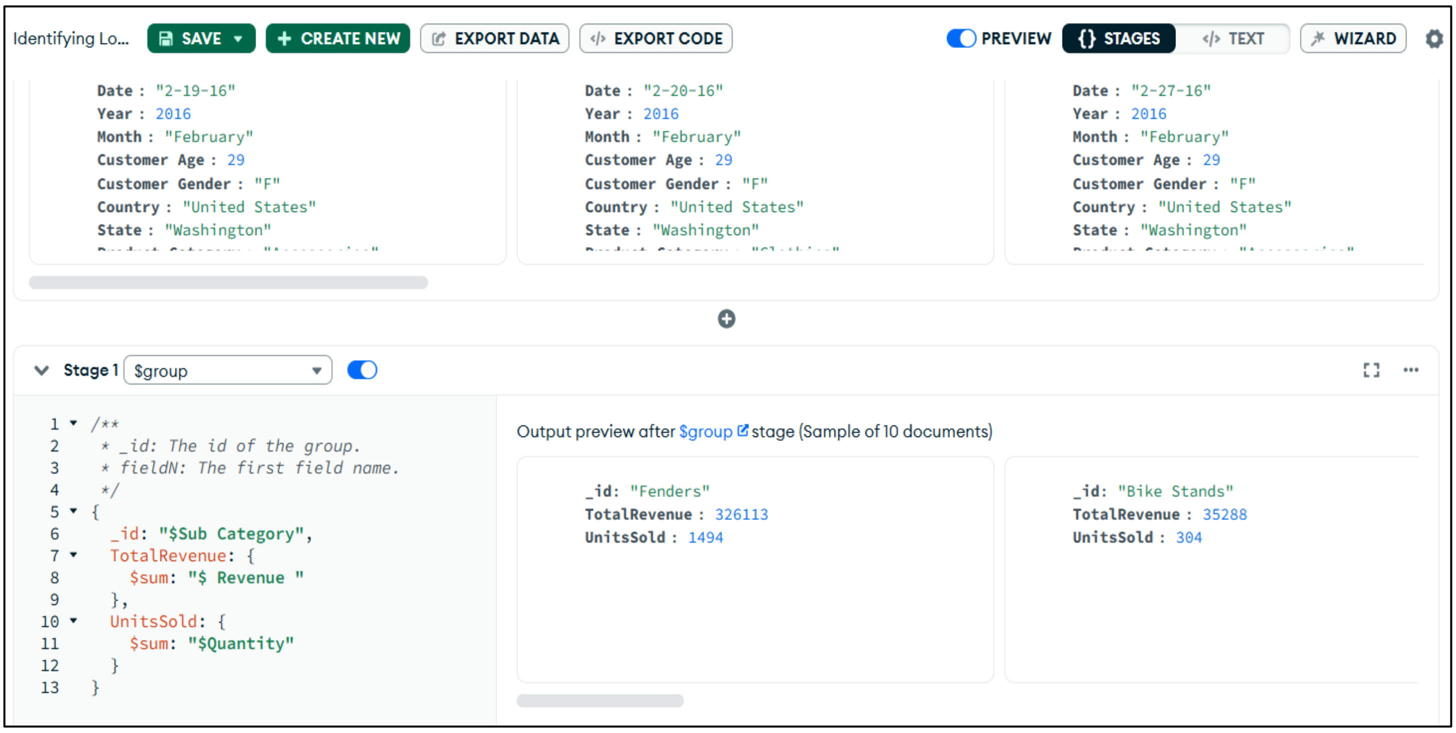
8. Customer spending Analysis by Age Group:

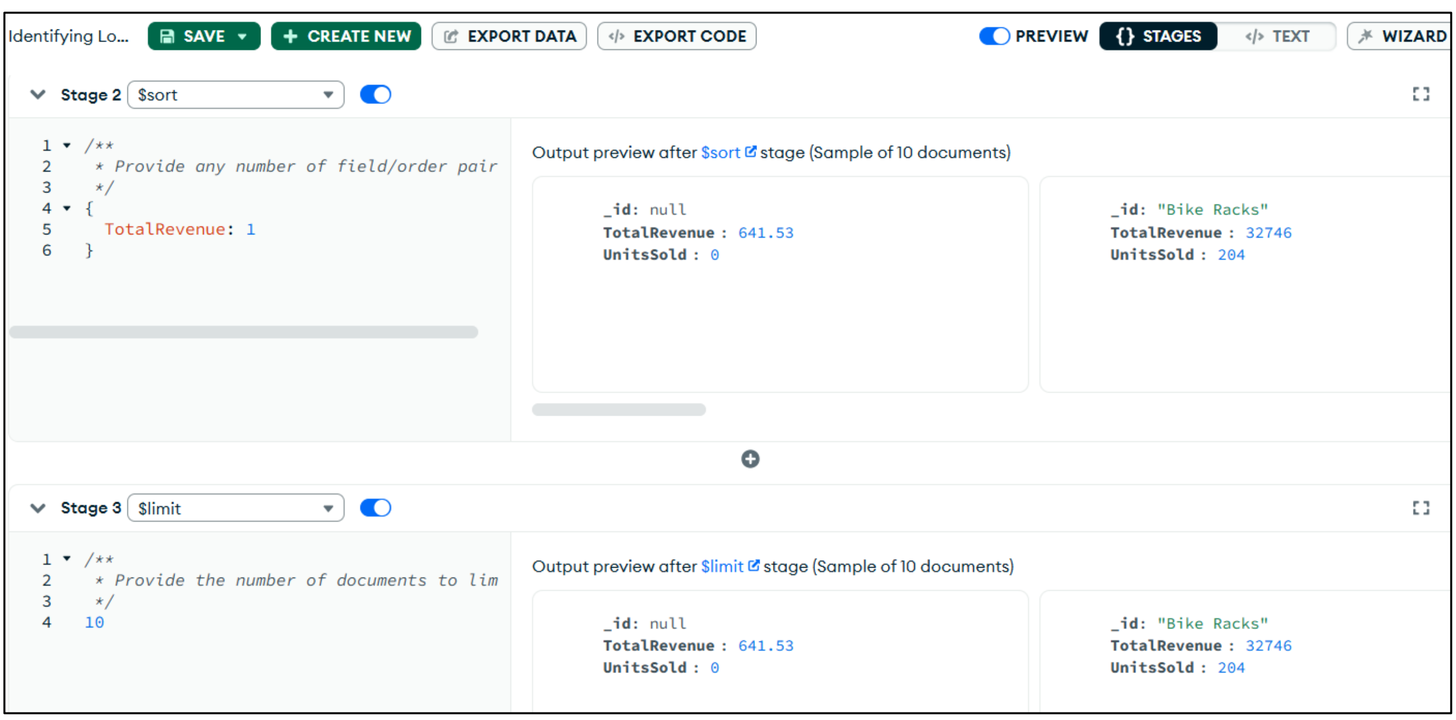
This demonstrates and displays the spending pattern of the customers present of every kind of age group, and shows the number of orders they have done, along with the total revenue it generated for the company in order to do the analysis.



09. Identifying low performing products:

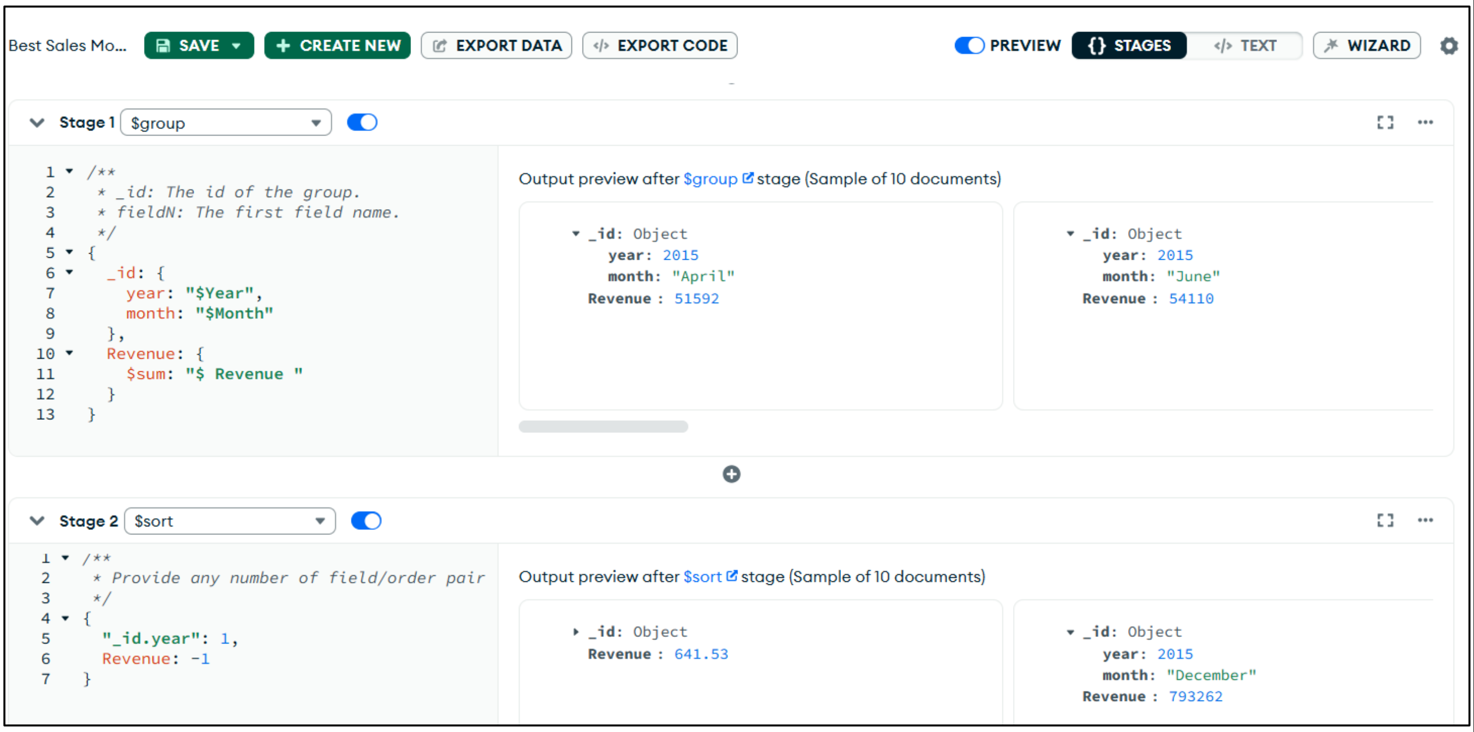
This analysis output shows the identification of low performing products from the dataset of 2015 and 2016 for the particular company, demonstrating the lowest performing products from using sub category and displays output as total revenue and units sold for the company.





10.Best Sales month by each year:

This demonstrates and displays the spending pattern of the customers present of every kind of age group, and shows the number of orders they have done, along with the total revenue it generated for the company in order to do the analysis.



DATA VISUALIZATION USING POWER BI

This is the overall content visualization of all the aggregation analysis we performed in the MongoDB. This also demonstrates clean and clear visuals of the analysis along with particular representation of the charts used for the visualization.